

R&I Tool for 797's OHT Steering Cylinder

Maintenance and Repair

Application

Administration of Components Life

Renewal of Component (CRC)

MARC Management

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1.0 Introduction

To ensure availability and reliability of equipment, it is necessary to include maintenance and repair tasks, which must be carried out efficiently, with quality, and safely. For large scale mining equipment, such as Caterpillar 797F trucks, these tasks include, among others, handling components of large size and weight. In order to achieve safety, efficiency and quality goals, the maintenance department must ensure that their staff is adequately trained, informed on support (both technical and procedures), provided with infrastructure, tools, and support equipment.

This Best Practice is concerned with the design, manufacture, use and benefits of a tool developed to replace and install (R&I) the steering cylinders on Caterpillar trucks so as to carry out a high standard of Quality and Safety work as well as save time and resources in the procedure.

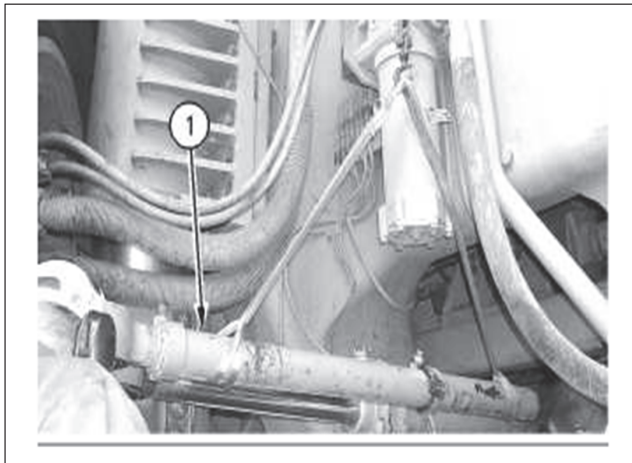
2.0 Description of Best Practice

2.1 Detection of Opportunity

The analysis of the R&I of steering cylinders on Caterpillar trucks highlighted four (4) major areas of improvement. These opportunities motivated the development of the proposed tool presented in this Best Practice.

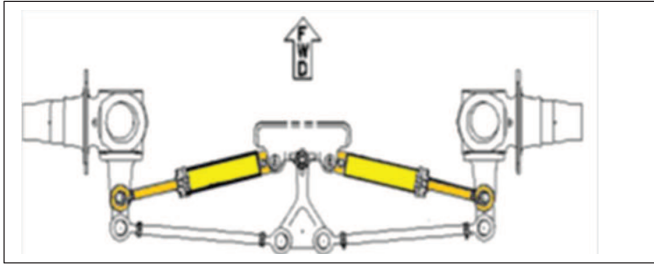
Areas of Opportunity

Safety



The current recommended procedure for cylinder replacement on Caterpillar trucks includes the use of slings, adjustment hoist(s), and a proper lifting equipment. This procedure is highly sensitive to the experience of maintenance staff. The technician(s) must maintain control and balance of the steering cylinder at all times, to avoid the risk of damage to the component, equipment, and more importantly causing personnel injuries. Working with suspended load is, in itself, a high-risk task.

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Simplified diagram of cylinder position on Truck

CAEX Model	Weight in Kilos	Weight in pound
777	63	140
793	110	243
795	318	700
797	329	725

Weight table for steering cylinders
on Caterpillar trucks

Time (Efficiency)

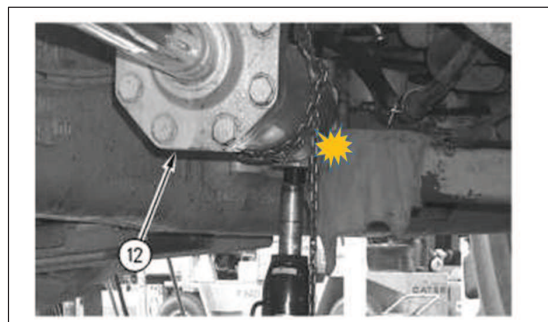
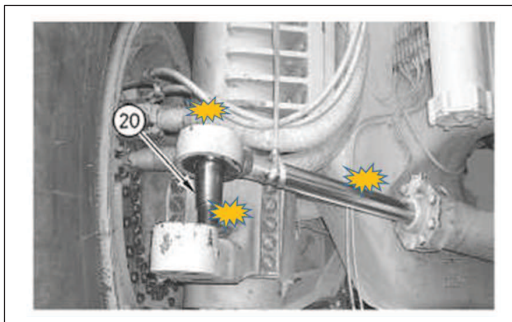
According to our records in the AMT repair logs, the time required for the task of R&I of the Steering Cylinder following the procedure suggested by Caterpillar, is in average 14 hours of machine down time. Additionally, it is worthy of mention that doing other jobs at the same time is not recommended when working with suspended load.

Resources (Labor)

Usually, the R&I work of Steering Cylinders on 797 Trucks is carried out by three (3) technicians. This means a total of 42 man-hours is required to carry out the task following the procedure suggested by Caterpillar.

Quality

During the replacement processes of Steering Cylinders, to safeguard the suspending load, it was observed that the technicians perform substandard maneuvers which may impact the quality of the job mainly due to the sensitivity required when handling the suspended load.

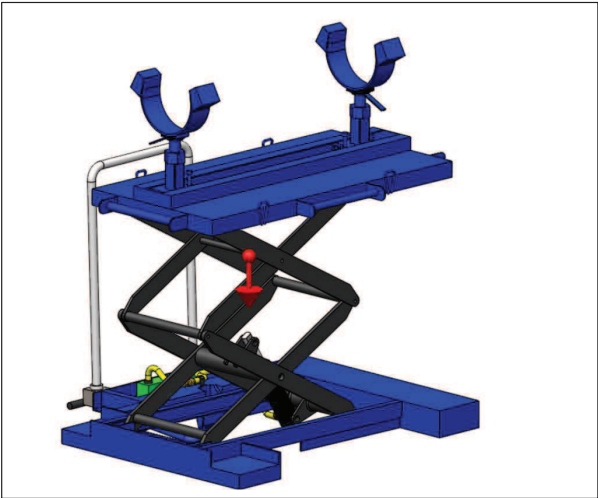


Impacted areas when handling suspended load

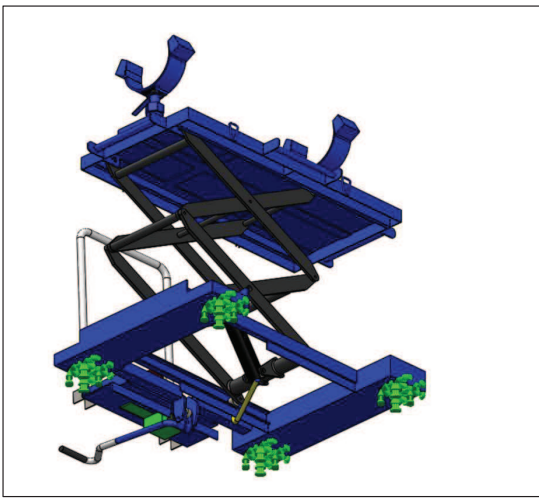
2.2 Proposed Improvement Solution

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The tool shown below has been designed to reduce the risk to which Support Technicians are exposed in the process of R&I Steering Cylinders on Caterpillar Trucks. Additionally, this tool simplifies the execution of the R&I task, impacting positively on the time and quality of it. It is worth to mention that the tool design makes it applicable to different truck models



Top view of Tool for Steering Cylinder Replacement



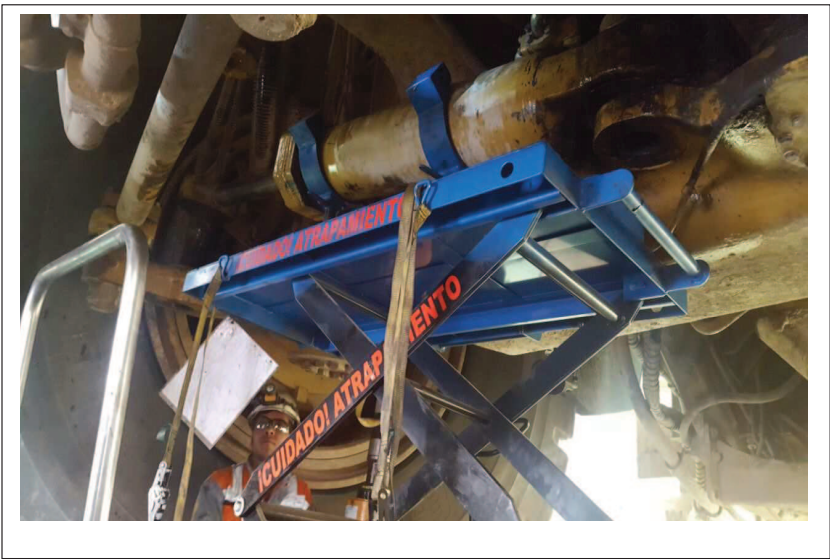
Bottom view of Tool for Steering Cylinder Replacement



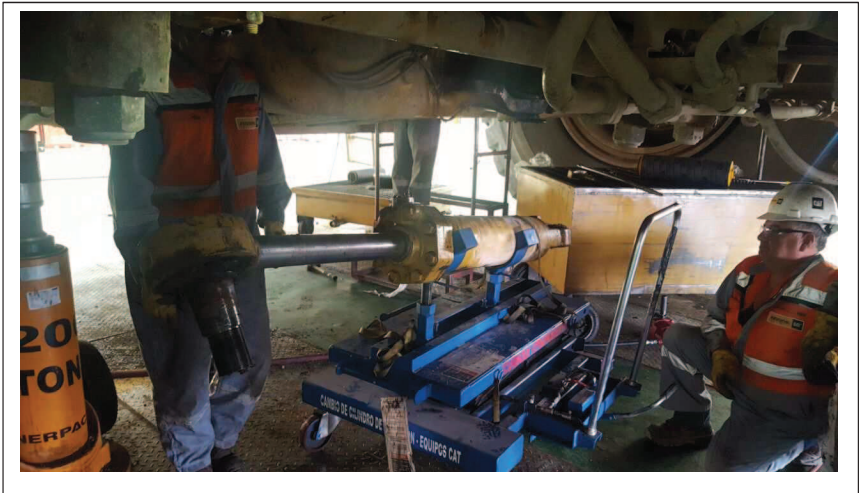
General view of Tool for Steering Cylinder Replacement

The following pictures show the Tool for Steering Cylinder replacement under operation at our workshops.

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Cylinder in place for installation or removal



Technicians operating the tool for cylinder replacement

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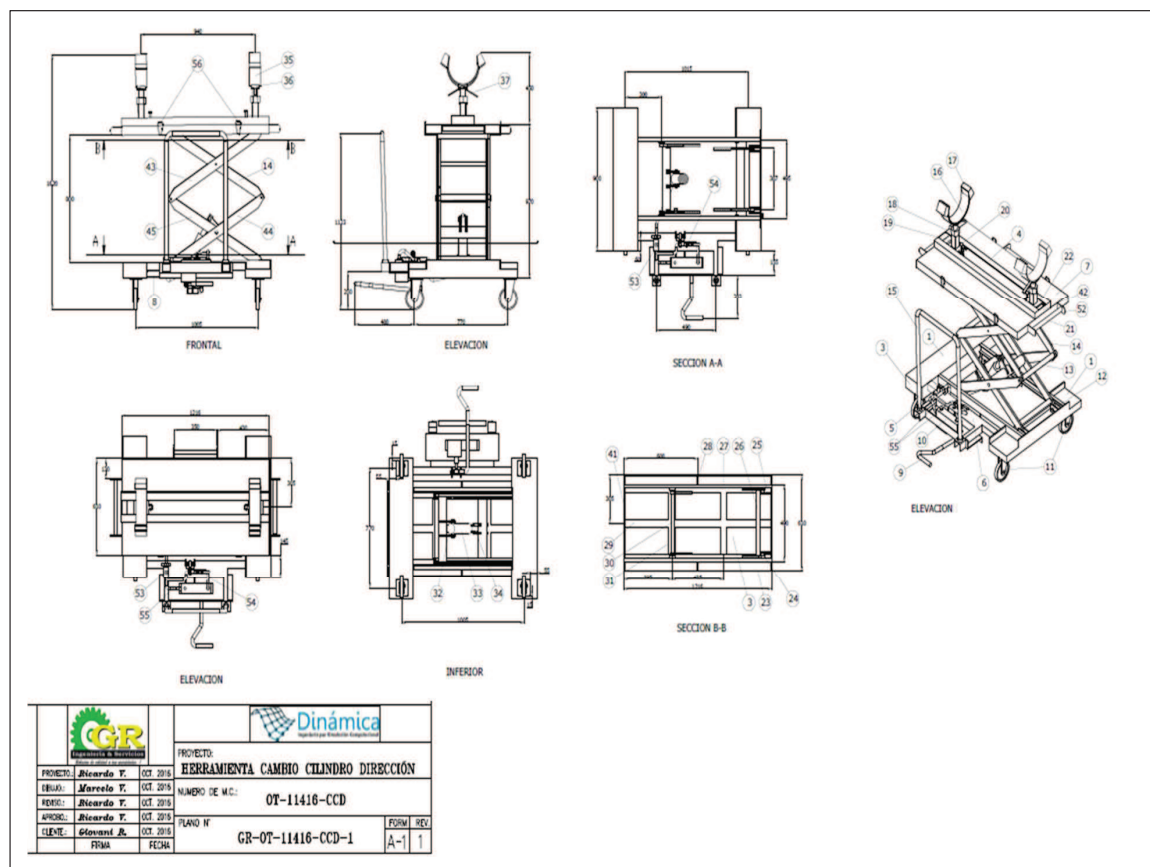
3.0 Steps for implementation

The general steps to follow when implementing this Best Practice are the following:

- 3.1 Manufacture of Tool for Steering Cylinder Replacement
- 3.2 Technical Certification of the Tool
- 3.3 Practical tests of replacement tasks of Steering Cylinders
- 3.4 Final adjustments
- 3.5 Documentation of SOP
- 3.6 Staff training
- 3.7 Operational start-up and Validation of Benefits

3.1 Manufacture of Tool for Steering Cylinder Replacement

The tool was manufactured locally according to the following design specifications. This design was created locally by Finning – DMH contract.



The manufacture diagram is attached along with this Best Practice document.

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Note: In our case the manufacture was carried out by the following local supplier:

GR Ingeniería & Servicios
 Av. Jerusalén No. 4166
 Antofagasta, Chile
 Jv.gringenieria@gmail.com

3.2 Technical certification of the Tool for R&I Accumulator Cylinder (Welding and Capacity)

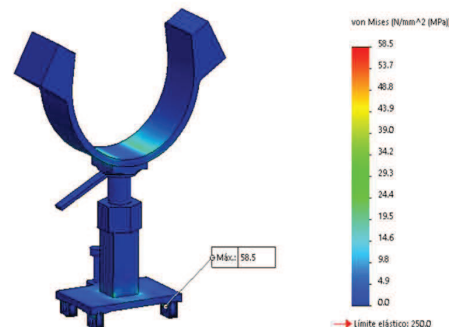
Once manufactured, the certification of the welding and capacity analysis of the accessory is relevant.



Structural Calculation Report and
 Quality Certification of
 Steering Cylinder

Example of tests / stress analysis performed on the Tool

U-shape support:
 Maximum Stress (Von Mises) = 58.5 MPa.
 Location of Maximum Stress = Wheel pin

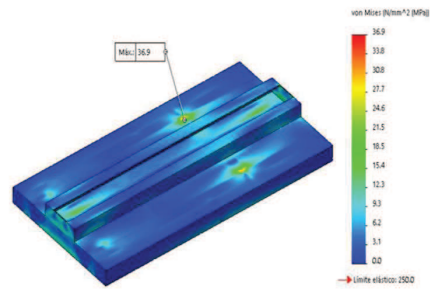


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Top tray:

Maximum Stress (Von Mises) = 37 MPa.

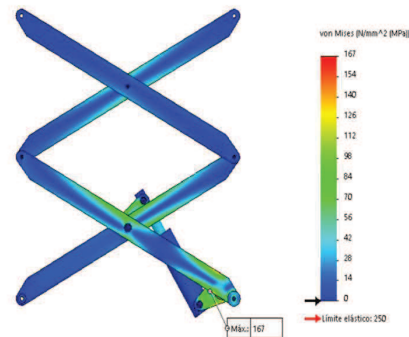
Location of Maximum Stress = Contact area with collapsible structure.



Collapsible structure:

Maximum Stress (Von Mises) = 167 MPa.

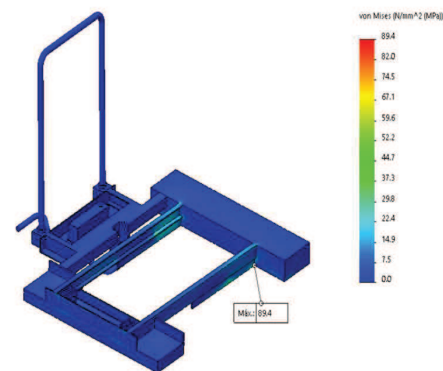
Location of Maximum Stress = Reticulated beam on bottom, wheel area.



Bottom Base:

Maximum Stress (Von Mises) = 89.4 MPa.

Location of Maximum Stress = reinforcement rib 5 mm.



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3.4 Practical tests on replacement tasks of Steering Cylinders.

Once the accessory is received, we recommend running operational tests at the workshop within a controlled and safe environment. This recommendation aims to identify and accomplish the final adjustments required.

3.5 Final adjustments

A prototype of this tool was used in a trial period in order to assess all the risk options both for users and components, as well as to identify design improvements for a more efficient performance. The observations found were fully corrected and its final design completely improved.

3.6 Documentation of SOP (Standard Operational Procedure)

In order to ensure the correct operation of this accessory, it is recommended to document the standard procedure of use adjusted to its operation (SOP).

Likewise, it is recommended to check the specific safety norms for its operation to adapt the SOP to the specific needs of the user.

3.7 Staff Training

It is important that the training be adequate for the staff who will be constantly using the tool in the tasks of Steering Cylinder replacement. At least 2 technicians should be available per shift.

It is recommended that the training includes the revision of the standard procedure (SOP) as well as practice of use within a safe and controlled environment.

3.8 Operational start-up and Validation of Benefits

Once the Tool for installation of the Steering Cylinder is available for operation, a follow-up must be implemented with the correct use and record of the benefits.

Once the expected benefits are obtained, the specific Standard Jobs must be improved / modified along with the department of Planning and Programming, especially the new shutdown times of machine and man-hours.

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4.0 Benefits

According to our observations and records of use for this tool in our operation, the benefits documented until now reflect the following improvements.

Benefit Area	Without Tool	With Tool	% of Benefit
Risk level	High	Controlled – Low	Reduced by 50%
Total shutdown time of machine	14	5	Reduced by 64%
Total Time in Hours Men needed	42	10	Reduced by 76%

As shown in the table above with results, the benefits are found in the areas of Safety, Performance Time of the Task, Use of Manpower, and Quality of Performance.

Safety: There is a change from suspended load to supported load, which significantly lowers the levels of risk.

Performance time (Shutdown Hours of Machine): The performance time of the task is reduced to 9 hours that is a 64 % reduction.

It is worthy of mention that the time reduction of the task is directly related to the MTTR result of the fleet. Its influence must be analyzed along with the replacement frequency of the Steering Cylinders.

Man-Hours (MH worked): Removing 1 support person from this task plus a machine-hour reduction causes a very significant impact with regards to maintenance ratio (MR). Man-hours are reduced by 76%, which results in 32 man-hours using this tool.

Quality: A work process involving lifting devices that hold a suspended load is of high risk both for the holders and the component held as there is a condition of sensitivity in handling the load. The negative consequences can imply damage to the component, which will later impact the work quality such as scratches on the chrome finish, loose connections, banged threads, etc. The use of this tool considerably reduces these risks as the load goes from being suspended to being supported, thus making mobility and maneuver easier for the holder. Such improvements will influence positively on the results of MTBS of the fleet.

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5.0 Resources Required

The resources required are those specified in this Best Practice.

6.0 Additional Support / References

The following information is attached to this Best Practice.

- Diagram / Manufacturing drawing

Note: The Structural Calculation Report and Structural Certification can be provided upon request.

7.0 Other related Best Practices

N/A

8.0 Acknowledgements

Creation, Implementation, and Documenting of the Best Practice

Fernando Valdés
Contract Manager
Contrato Ministro Hales (Codelco Chile)
Mobile.: +56 984180500
E-mail: fernando.valdes@finning.com

Marcos Palma Albornoz
Head of the Continuous Improvement Dept.
Contrato Ministro Hales (Codelco Chile)
Mobile.: +56 961254692
E-mail: marcos.palma@finning.com

Best Practice written by:

Juan Crooker Acosta
Coordinator of Best Practices Program
Finning SA – Antofagasta
Mobile: +56 966163251
E-mail: juan.crooker@finning.com

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